

CStats - Lab #4

March 17, 2026

Exercise 1.

A computer ANOVA output is shown in Table 1. Fill in the blanks. You may give bounds on the p -value.

One-way ANOVA					
Source	DF	SS	MS	F	p
Factor	3	36.15	?	?	?
Error	?	?	?		
Total	19	196.04			

Table 1: One-way ANOVA Table

Solution 1.

$$\text{Error DF} = \text{Total DF} - \text{Factor DF} = 19 - 3 = 16$$

Let calculate sum of squares (SS).

$$\text{Error SS} = \text{Total SS} - \text{Factor SS} = 196.04 - 36.15 = 159.89$$

Then, the mean square (MS) is computed as $MS = \frac{SS}{DF}$.

$$\text{Factor MS} = \frac{36.15}{3} = 12.05$$

$$\text{Error MS} = \frac{159.89}{16} = 9.993$$

On the other hand, the F -statistic is given by

$$F = \frac{\text{Factor MS}}{\text{Error MS}} = \frac{12.05}{9.993} \approx 1.206.$$

For DF Factor = 3 and DF Error = 16, we check the F -distribution table.

$$F_{0.10} = 2.46, \quad F_{0.05} = 3.24, \quad F_{0.025} = 4.07, \quad F_{0.01} = 5.29$$

Since our computed $F = 1.206$ is much smaller than 2.46, the p -value is greater than 0.10, which is $p > 0.10$.

Exercise 2.

The engineer is interested in determining if the RF power setting affects the etch rate, and she has run a completely randomized experiment with four levels of RF power and five replicates. The data are shown in Table 2. Construct the ANOVA table and test the null hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$, for the significance level $\alpha = 0.05$.

RF Power (W)	Observed Etch Rate ($\text{\AA}/\text{min}$)					Totals	Averages
	1	2	3	4	5	$y_{i.}$	$\bar{y}_{i.}$
160	575	542	530	539	570	2756	551.2
180	565	593	590	579	610	2937	587.4
200	600	651	610	637	629	3127	625.4
220	725	700	715	685	710	3535	707.0
						$y_{..} = 12,355$	$\bar{y}_{..} = 617.75$

Table 2: The Plasma Etching Experiment

Solution 2.

One-way ANOVA Table

Source	SS	DF	MS	F
Factor	66870.55	3	22290.18	66.80
Error	5339.20	16	333.70	
Total	72209.75	19		

Using F-table, we have $F_{0.05}(3, 16) = 3.24$. Since $66.80 > 3.24$, we reject H_0 and conclude that the RF power setting significantly affects the mean etch rate.

Exercise 3.

Let X_1, X_2, X_3, X_4 equal the cholesterol level of a woman under the age of 50, a man under 50, a woman 50 or older, and a man 50 or older, respectively. Assume that the distribution of X_i is $N(\mu_i, \sigma^2)$, $i = 1, 2, 3, 4$.

(a) Define the null hypothesis and give a critical region for an $\alpha = 0.05$ significance level.

(b) Construct an ANOVA table and state your conclusion, using the following data:

X_1 : 221 213 202 183 185 197 162

X_2 : 271 192 189 209 227 236 142

X_3 : 262 193 224 201 161 178 265

X_4 : 192 253 248 278 232 267 289

(c) Give bounds on the p -value for this test.

Solution 3.

(a) Since we want to find whether there is any difference in cholesterol levels between different groups, using seven observations, we are testing the null hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$. We can use an ANOVA F -test, where the test statistic follows an F -distribution with degrees of freedom such that

$$df_A = a - 1 = 4 - 1 = 3$$

$$df_E = a.r - a = 28 - 4 = 24 \quad \text{Note: total degrees of freedom is 27 (28-1)}$$

where k is the number of groups and N is the number of observations. From an F -table, the critical value at $\alpha = 0.05$ is $F_{0.05,(3,24)} = 3.01$. Thus, the critical region is $F > 3.01$.

(b) To construct an ANOVA table, we need to find degrees of freedom (DF), sum of squares (SS), mean squares (MS) and F -test. Factor refers to the categorical independent variables whose effect on the dependent variable we are analyzing. Sum of Squares Treatments (SSA) measures how much the group means differ from the overall mean. On the other hand, Error represents variability within each group. Sum of Squares Error (SSE) measures how much individual observations within a group deviate from their group mean. The sample mean for each group and overall mean are calculated by the followings:

$$\bar{X}_1 = \frac{221 + 213 + 202 + 183 + 185 + 197 + 162}{7} = 194.71$$

$$\bar{X}_2 = \frac{271 + 192 + 189 + 209 + 227 + 236 + 142}{7} = 209.43$$

$$\bar{X}_3 = \frac{262 + 193 + 224 + 201 + 161 + 178 + 265}{7} = 212.00$$

$$\bar{X}_4 = \frac{192 + 253 + 248 + 278 + 232 + 267 + 289}{7} = 251.29$$

$$\bar{X} = \frac{7(194.71 + 209.43 + 212.00 + 251.29)}{28} = 216.86 \quad \text{Overall Mean}$$

Then, we can calculate the sum of squares SSA and SSE as

$$\begin{aligned}
 SSA &= 7 \sum (\bar{X}_i - \bar{X})^2 = 7((194.71 - 216.86)^2 + (209.43 - 216.86)^2 \\
 &\quad + (212.00 - 216.86)^2 + (251.29 - 216.86)^2) \\
 &= 7((-22.15)^2 + (-7.43)^2 + (-4.86)^2 + (34.43)^2) \\
 &= 7(490.56 + 55.22 + 23.62 + 1185.71) = 7 \cdot 1755.11 = 12285.77
 \end{aligned}$$

$$\begin{aligned}
 SSE &= \sum \sum (X_{ij} - \bar{X}_i)^2 \\
 SSE_1 &= 1308.86 \\
 SSE_2 &= 3264.86 \\
 SSE_3 &= 4148.00 \\
 SSE_4 &= 3553.43 \\
 SSE &= 1308.86 + 3264.86 + 4148.00 + 3553.43 = 12275.14
 \end{aligned}$$

Also, let compute mean squares between groups (MSB) and mean squares within-groups (MSW) by the followings:

$$\begin{aligned}
 MSA &= \frac{SSA}{df_A} = \frac{12285.77}{3} = 4095.26 \\
 MSE &= \frac{SSE}{df_E} = \frac{12275.14}{24} = 511.46
 \end{aligned}$$

Then, we are ready to apply F -test.

$$F = \frac{MSA}{MSE} = \frac{4095.26}{511.46} = 8.01$$

One-way ANOVA Table

Source	SS	DF	MS	F
Factor	12285.77	3	4095.26	8.01
Error	12275.14	24	511.46	
Total	24560.91	27		

Since $F = 8.01 > 3.01$, we reject H_0 at the $\alpha = 0.05$ significance level, which means that at least one group has a different mean cholesterol level.

(c) From F -table, we have $F_{0.005,(3,24)} = 5.52$. Since $8.01 > 5.52$, the p -value is less than 0.005, therefore, $p < 0.005$.

Exercise 4.

A pharmaceutical manufacturer wants to investigate the bioactivity of a new drug. A completely randomized single-factor experiment was conducted with three dosage levels, and the following results were obtained:

Dosage	Observations
20g	24 28 37 30
30g	37 41 31 35
40g	42 47 52 38

- (a) Is there evidence to indicate that dosage level affects bioactivity? Use $\alpha = 0.05$.
 (b) Analyze the residuals from this experiment and comment on model adequacy.

Solution 4.

(a) The null hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3$ (all mean bioactivities are equal) against the alternative hypothesis at least one mean differs. We have one factor ANOVA table as follows:

One-way ANOVA Table					
Source	SS	DF	MS	F	$p - value$
Factor	454.17	2	227.08	8.13	0.009632
Error	251.50	9	27.94		
Total	705.67	11			

From F-table, $F_{0.05}(2, 9) \approx 4.26$. Since $8.13 > 4.26$, we reject H_0 . On the other hand, since the calculated p-value is less than the given significance level, we reject H_0 . It means that There is significant evidence that dosage level affects bioactivity.

(b) We need to evaluate whether the ANOVA model assumptions are satisfied using the residuals. We check three main assumptions:

1. Independent errors
2. Normally distributed errors
3. Constant standard deviation of errors

Residuals are calculated by

$$e_{ij} = y_{ij} - \bar{y}_i$$

Thus,

20g	-5.75, -1.75, 7.25, 0.25
30g	1, 5, -5, -1
40g	-2.75, 2.25, 7.25, -6.75

1) Residuals should show no systematic pattern. Since the experiment is described as a completely randomized design, the independence assumption is likely satisfied.

2) Residuals should behave like random noise around zero if the model is appropriate. A normal probability plot is given in Figure 1. What we check are residuals should lie approximately on a straight line and no extreme outliers. Thus, we observe that

- Residuals range roughly from -6.75 to 7.25
- They are fairly symmetrically distributed around 0
- Both positive and negative values appear in each group
- Consequently, there is no strong indication that residuals deviate from normality.

3) We check whether the spread of residuals is similar across dosage levels. We have the following residual ranges:

20 g	-5.75 to 7.25
30 g	-5 to 5
40 g	-6.75 to 7.25

We observe that all groups have similar magnitude of variation and residuals are balanced above and below zero, as shown in Figure 2. Thus, we conclude with that the constant variance assumption appears reasonable.

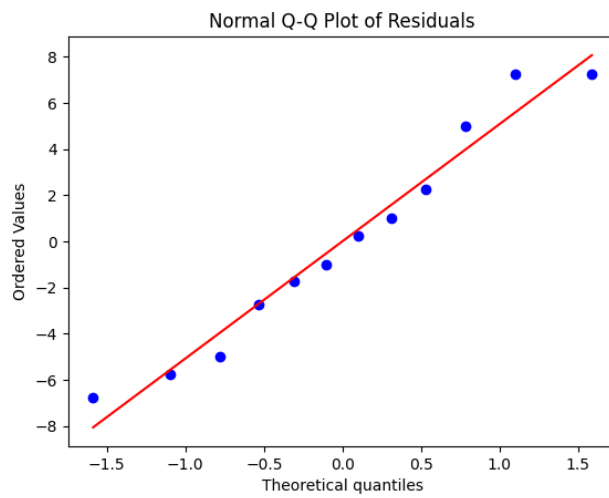


Figure 1: Residual Plot 1

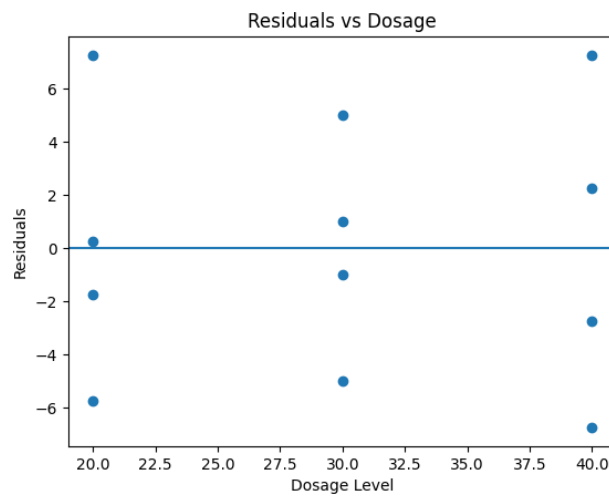


Figure 2: Residual Plot 2